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EE25BTECH11057 - Rushil Shanmukha Srinivas

Question : The area between $x = y^2$ and $x = 4$ is divided into two equal parts by the line $x = a$, find the value of a .

Solution :

Name	Value
Parabola	$\mathbf{x}^\top \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} - \begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{x} = 0$
Line1	$\mathbf{x} = \begin{pmatrix} a \\ 0 \end{pmatrix} + \kappa_1 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$
Line2	$\mathbf{x} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} + \kappa_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Table : Conics

On comparing Parabola with

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \quad (0.1)$$

The parameters of the Parabola are

$$\mathbf{V} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \mathbf{u} = \frac{-1}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, f = 0 \quad (0.2)$$

For the line $x - a = 0$, the parameters are

$$\mathbf{h}_1 = \begin{pmatrix} a \\ 0 \end{pmatrix}, \mathbf{m}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (0.3)$$

For the line $x - 4 = 0$, the parameters are

$$\mathbf{h}_2 = \begin{pmatrix} 4 \\ 0 \end{pmatrix}, \mathbf{m}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (0.4)$$

The points of intersection for a conic and a line is given by

$$\kappa_i = \frac{1}{\mathbf{m}^\top \mathbf{V} \mathbf{m}} \left(-\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) \pm \sqrt{[\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u})]^2 - g(\mathbf{h}) (\mathbf{m}^\top \mathbf{V} \mathbf{m})} \right) \quad (0.5)$$

$$g(\mathbf{x}) = \mathbf{x}^\top \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} - \begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{x} \quad (0.6)$$

$$g(\mathbf{h}) = \mathbf{h}^\top \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{h} - \begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{h} \quad (0.7)$$

$$\kappa_i = \left(0 \pm \sqrt{\begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{h} - \mathbf{h}^\top \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{h}} \right) \quad (0.8)$$

So

$$\kappa_1 = \sqrt{a}, -\sqrt{a} \text{ and } \kappa_2 = 2, -2 \quad (0.9)$$

Therefore the points of intersection are

For line 1

$$\mathbf{P}_1 = \begin{pmatrix} a \\ \sqrt{a} \end{pmatrix}, \mathbf{P}_2 = \begin{pmatrix} a \\ -\sqrt{a} \end{pmatrix} \quad (0.10)$$

For line 2

$$\mathbf{P}_3 = \begin{pmatrix} 4 \\ 2 \end{pmatrix}, \mathbf{P}_4 = \begin{pmatrix} 4 \\ -2 \end{pmatrix} \quad (0.11)$$

The area of the parabola in between the lines $x = a$ and $x = 4$ is given by

$$2 \left(\int_0^4 \sqrt{x} dx - \int_0^a \sqrt{x} dx \right) = \frac{4}{3} \left(8 - a^{\frac{3}{2}} \right) \quad (0.12)$$

The area between parabola and line $x = a$ is given by

$$2 \left(\int_0^a \sqrt{x} dx \right) = \frac{4}{3} a^{\frac{3}{2}} \quad (0.13)$$

Given both the areas are equal

$$\frac{4}{3} \left(8 - a^{\frac{3}{2}} \right) = \frac{4}{3} a^{\frac{3}{2}} \quad (0.14)$$

$$4 = a^{\frac{3}{2}} \quad (0.15)$$

$$a = 4^{\frac{2}{3}} = 16^{\frac{1}{3}} \quad (0.16)$$

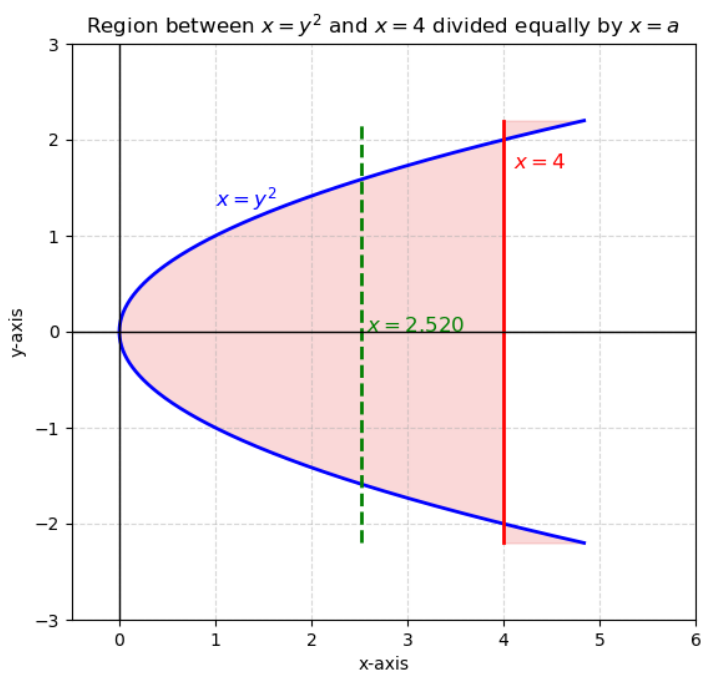


Fig : Parabola and Lines